

Rigid Anisotropic Gaussian Model for the 12 CAIN Test Particles

OPALX Beam–Beam Sandbox

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1 Purpose

This note documents the manufactured tracking model implemented in the `track12particles.py` sandbox script. The goal is to compare the uploaded CAIN test-particle trajectories against a reproducible continuum model for the beam–beam force. The model is deliberately simple: the source bunch is a rigid anisotropic Gaussian moving at constant velocity, with an optional finite-macroparticle centroid fluctuation. It is not a PIC model and does not include bunch disruption, higher-order macro-particle noise, or self-consistent source evolution.

The CAIN reference file contains twelve right-going test particles: six electron–positron pairs injected at

$$ct/\sigma_z = -1.5, -1.0, -0.5, 0, 0.5, 1.0, \quad \sigma_z = 0.6 \text{ mm}.$$

The plotted comparison uses s in millimetres and x in micrometres. Spin columns are parsed but are not part of the present comparison.

2 Source Selection Convention

The script supports three source selections:

- **oncoming**: only the electron bunch moving opposite to the right-going test particles is included;
- **copropagating**: only the electron bunch moving in the same direction as the test particles is included;
- **both**: two equal electron bunches colliding head-on are included.

The current default for the 12-particle comparison is **oncoming**. This does not mean that both bunches collide head-on. The head-on two-bunch model is **both**. The **oncoming** choice is useful because the dominant force on a right-going ultra-relativistic witness comes from the opposite-going source; for a co-propagating ultra-relativistic source, the electric and magnetic parts nearly cancel in the Lorentz force.

3 Rigid Anisotropic Gaussian Source

The source is specified by lab-frame beam sizes

$$\sigma_x = \sigma_y = 1.944\,325\,075\,701 \text{ }\mu\text{m}, \quad \sigma_z = 0.6 \text{ mm},$$

and by a source kinetic energy $K_s = 245 \text{ MeV}$. The source speed is

$$\gamma_s = 1 + \frac{K_s}{m_e c^2}, \quad \beta_s = \sqrt{1 - \gamma_s^{-2}}.$$

For a source moving along $\pm z$, the bunch centre is

$$\mathbf{r}_c(t) = \mathbf{r}_c(t_0) + \boldsymbol{\beta}_s c(t - t_0), \quad \boldsymbol{\beta}_s = (0, 0, \pm\beta_s).$$

The transverse sizes are unchanged by the boost. The rest-frame longitudinal size is dilated:

$$\sigma'_x = \sigma_x, \quad \sigma'_y = \sigma_y, \quad \sigma'_z = \gamma_s \sigma_z.$$

In the source rest frame the charge density is

$$\rho'(\mathbf{r}') = \frac{Q}{(2\pi)^{3/2} \sigma'_x \sigma'_y \sigma'_z} \exp \left[-\frac{1}{2} \left(\frac{x'^2}{\sigma_x'^2} + \frac{y'^2}{\sigma_y'^2} + \frac{z'^2}{\sigma_z'^2} \right) \right],$$

with $Q = -N_e e$, $N_e = 1.25 \times 10^{10}$.

4 Rest-Frame Field Integral

At a lab event $(t, \mathbf{r})$, the displacement from the moving source centre is split into transverse and longitudinal components. For the present straight-line geometry this is simply

$$x' = x - x_c(t), \quad y' = y - y_c(t), \quad z' = \gamma_s [z - z_c(t)].$$

The rest-frame electric field of the triaxial Gaussian is evaluated with the one-dimensional ellipsoidal integral

$$E'_i(\mathbf{r}') = \frac{Q x'_i}{4\pi\epsilon_0 \sqrt{2\pi}} \int_0^\infty \frac{\exp \left[-\frac{1}{2} \sum_j \frac{x_j'^2}{\sigma_j'^2 + u} \right]}{(\sigma_i'^2 + u) \sqrt{(\sigma_x'^2 + u)(\sigma_y'^2 + u)(\sigma_z'^2 + u)}} du.$$

The script evaluates this integral by trapezoidal quadrature in $\log u$. This is important for the present aspect ratio: σ'_z/σ'_x is large, so a linear quadrature can miss the transverse variance scale and underestimate the field by orders of magnitude.

5 Boost to Lab Frame

The rest-frame magnetic field is zero. For a boost along z , the lab fields are

$$E_x = \gamma_s E'_x, \quad E_y = \gamma_s E'_y, \quad E_z = E'_z,$$

and

$$\mathbf{B} = \frac{1}{c^2} \mathbf{v}_s \times \mathbf{E}.$$

The magnetic field is retained in the particle pusher. In the present comparison typical peak fields are in the GV/m range and $\mathbf{B}$ is of order 10–20 T near overlap.

6 Witness Push

The witness momentum is represented in OPALX-normalized units

$$\mathbf{P} = \gamma_w \boldsymbol{\beta}_w.$$

At each substep the script applies the same Boris kick convention used in the existing manufactured witness prototype:

$$\mathbf{P}^- = \mathbf{P}^n + \frac{1}{2} \Delta t \frac{qc}{m_e c^2} \mathbf{E}^n, \quad (1)$$

$$\mathbf{t} = \frac{1}{2} \Delta t \frac{qc^2}{\gamma^- m_e c^2} \mathbf{B}^n, \quad \mathbf{s} = \frac{2\mathbf{t}}{1 + \mathbf{t} \cdot \mathbf{t}}, \quad (2)$$

$$\mathbf{P}' = \mathbf{P}^- + \mathbf{P}^- \times \mathbf{t}, \quad (3)$$

$$\mathbf{P}^+ = \mathbf{P}' + \mathbf{P}' \times \mathbf{s}, \quad (4)$$

$$\mathbf{P}^{n+1} = \mathbf{P}^+ + \frac{1}{2} \Delta t \frac{qc}{m_e c^2} \mathbf{E}^n. \quad (5)$$

The position is advanced with a trapezoidal drift using the velocities before and after the kick:

$$\mathbf{r}^{n+1} = \mathbf{r}^n + \frac{1}{2} [\mathbf{v}(\mathbf{P}^n) + \mathbf{v}(\mathbf{P}^{n+1})] \Delta t.$$

7 Timestep Handling

The CAIN file provides output samples on a grid in ct . The manufactured tracker uses the same output grid for comparison, but internally substeps each output interval. The default maximum substep is

$$\Delta t_{\max} = 1 \text{ fs.}$$

This is intentionally small compared with the overlap transient, and it avoids confusing output cadence with integration cadence. The maximum substep can be changed with the `max-substep-s` option.

8 Finite-Macroparticle Centroid Model

CAIN represents each bunch by a finite random macroparticle sample. The manufactured model includes a simple leading-order surrogate for this effect: the rigid Gaussian source centroid is displaced by a reproducible random offset with rms

$$\sigma_{\bar{x}_i} = \frac{\sigma_i}{\sqrt{N_{\text{MC}}}},$$

where N_{MC} is the number of source macroparticles. The default used here is

$$N_{\text{MC}} = 400000.$$

For the transverse beam size this gives an rms centroid fluctuation of about

$$\frac{1.944 \mu\text{m}}{\sqrt{400000}} \simeq 3.1 \text{ nm.}$$

The random offset is generated from a fixed seed, so the manufactured trajectory is reproducible. This breaks the exact $y = 0$ symmetry and produces a nonzero vertical field, but it is only a centroid-noise model. It does not represent full PIC shot noise, local density fluctuations, or self-consistent beam disruption.

9 Current Comparison and Interpretation

The default command is

```
python sandbox/track12particles/track12particles.py \
  --input /Users/adelman/Desktop/TestParticleOrbit.dat
```

and writes the CAIN/reference plot, the manufactured-only plot, and the side-by-side comparison under `sandbox/track12particles/outputs`.

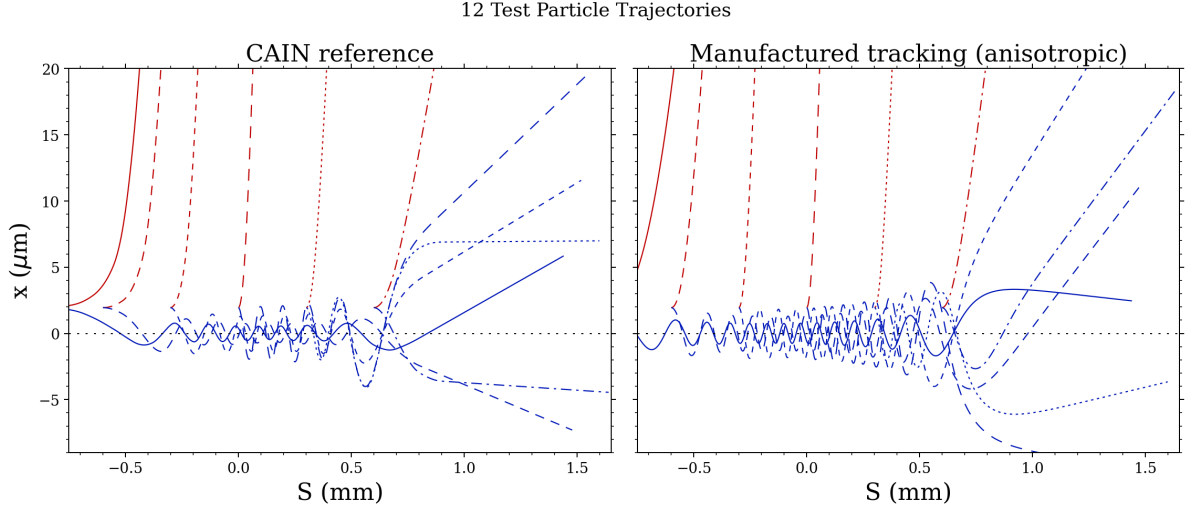


Figure 1: CAIN reference trajectories compared with the rigid anisotropic Gaussian manufactured model for $x(s)$, using the default `oncoming` source selection.

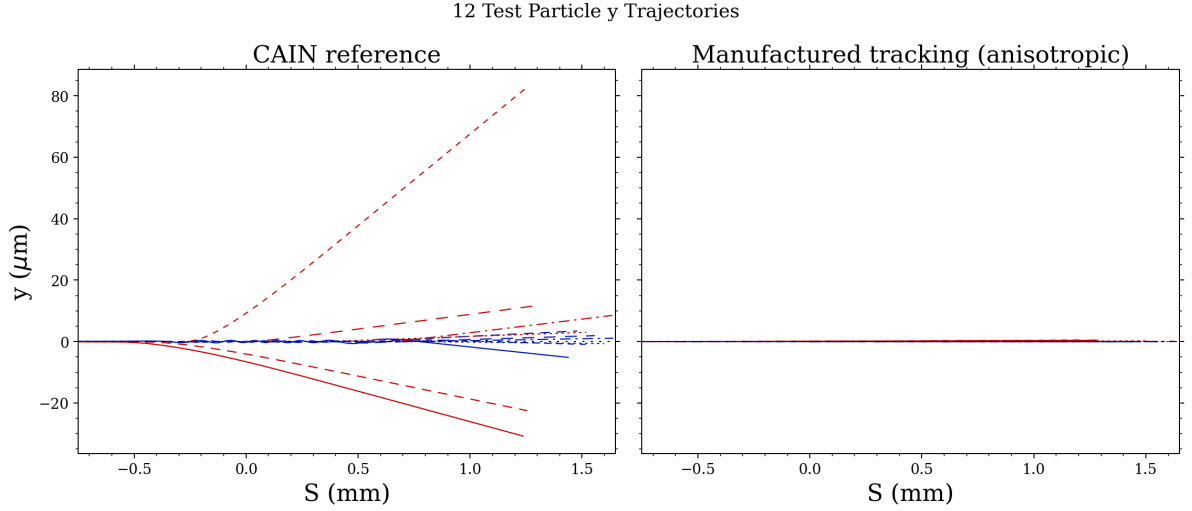


Figure 2: CAIN reference trajectories compared with the rigid anisotropic Gaussian manufactured model for $y(s)$, using the default `oncoming` source selection and the $1/\sqrt{N_{MC}}$ centroid fluctuation. The induced manufactured y motion is much smaller than the largest CAIN out-of-plane excursions for the seed used here.

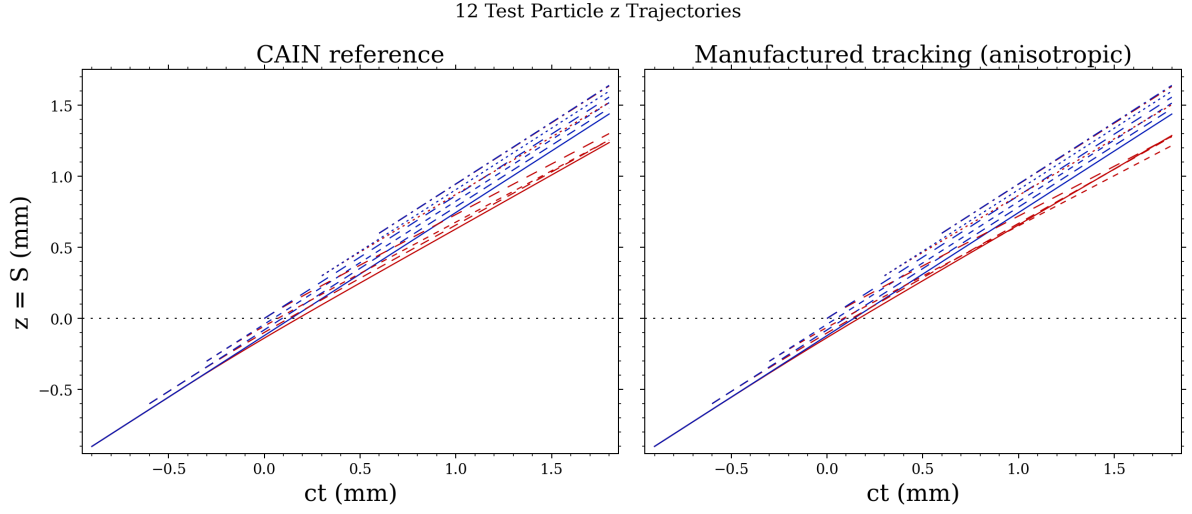


Figure 3: CAIN reference trajectories compared with the rigid anisotropic Gaussian manufactured model for longitudinal motion, plotted as $z = S$ versus ct , using the default `oncoming` source selection.

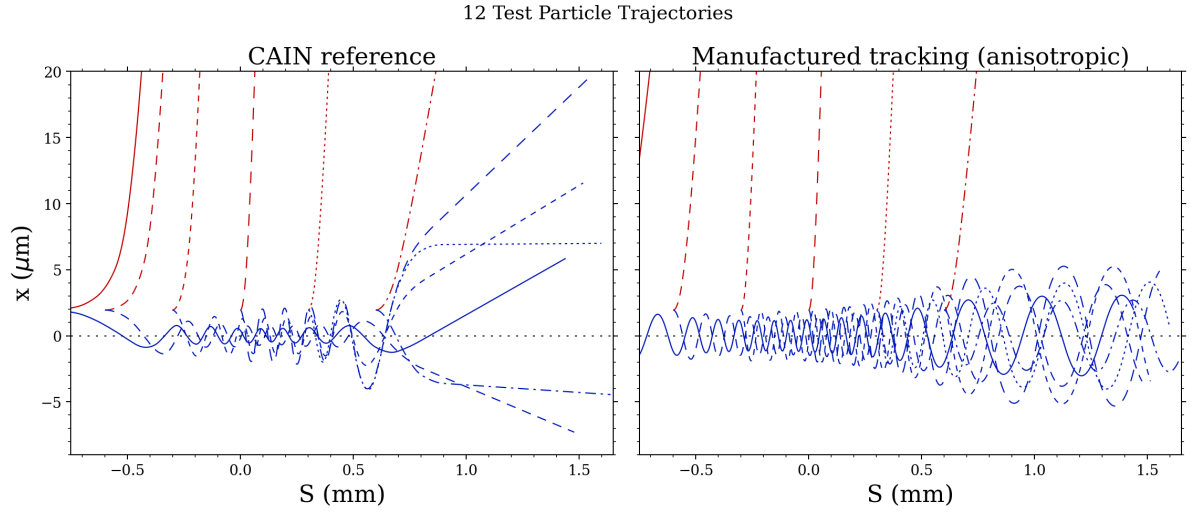


Figure 4: CAIN reference trajectories compared with the same rigid anisotropic Gaussian model for $x(s)$, using `--source-selection both`, i.e. two equal head-on electron bunches.

12 Test Particle y Trajectories

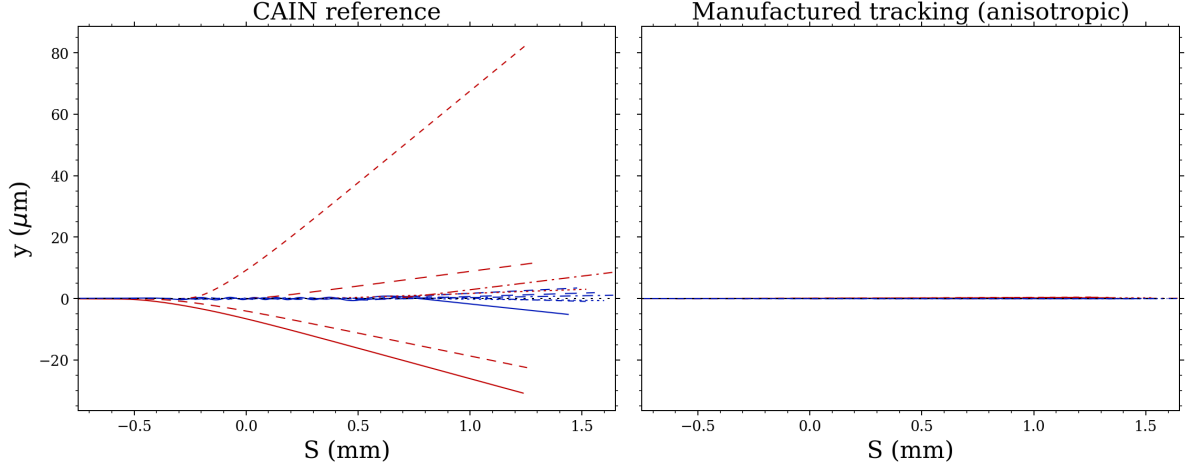


Figure 5: CAIN reference trajectories compared with the rigid anisotropic Gaussian manufactured model for $y(s)$, using `--source-selection` both and the same finite-macroparticle centroid fluctuation model.

12 Test Particle z Trajectories

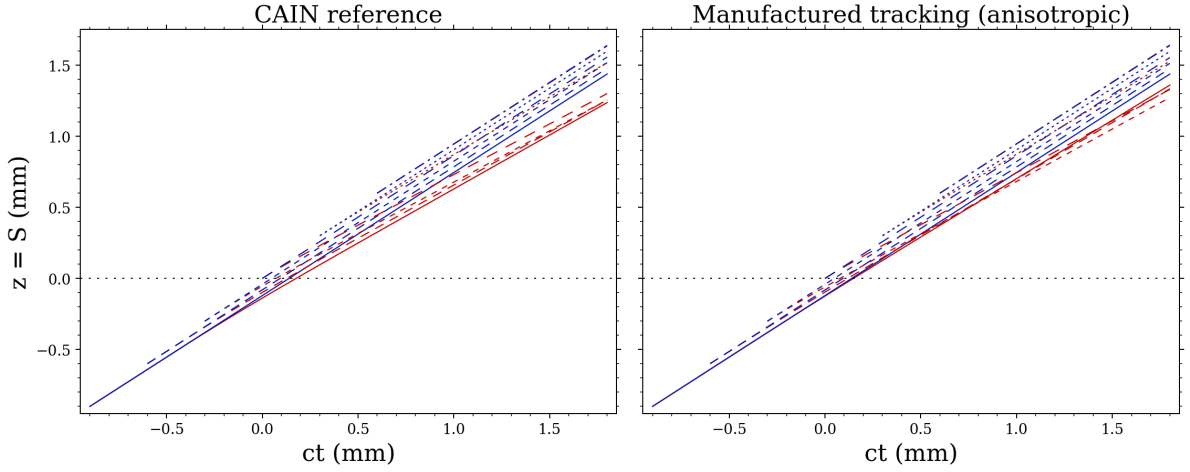


Figure 6: CAIN reference trajectories compared with the rigid anisotropic Gaussian manufactured model for longitudinal motion, plotted as $z = S$ versus ct , using `--source-selection` both.

10 OPALX–Manufactured Initial Conditions

10.1 Timing Convention from the Source Slides

The PowerPoint file `TestParticleOrbitSimulation.pptx` defines the test particles as artificial insertions during the simulation, not as Breit–Wheeler outputs. The relevant timing statement is:

$$ct/\sigma_z = -1.5, -1.0, -0.5, 0, 0.5, 1.0,$$

with the explicit interpretation that $ct/\sigma_z = -1.5$ is the moment when the test particle touches the oncoming bunch edge. The same slide states that the test particles are right-going. Together with the source-slide statement that the bunch longitudinal Gaussian is cut at $3\sigma_z$, this fixes the natural source timing convention.

If the oncoming primary electron bunch centre is at the IP at $ct = 0$, then for a rigid oncoming source moving toward decreasing s ,

$$s_c(ct) = -ct.$$

The initial test-particle coordinates satisfy $s_i = ct_i$. Therefore, at the first timing point $ct_i = -1.5\sigma_z$, the source centre is at $+1.5\sigma_z$, and the test particle is separated from it by $-3\sigma_z$, exactly the stated bunch edge. Thus the primary bunch should be sampled or evaluated at the test-particle insertion time $t_i = ct_i/c$, not at one common $t = 0$ event for all twelve witnesses.

Pair	ct_i/σ_z	ct_i [mm]	oncoming $s_c(ct_i)$ [mm]	$(s_i - s_c)/\sigma_z$
1	-1.500	-0.9000	0.9000	-3.000
2	-0.999	-0.5994	0.5994	-1.998
3	-0.501	-0.3006	0.3006	-1.002
4	0.000	0.0000	0.0000	0.000
5	0.501	0.3006	-0.3006	1.002
6	0.999	0.5994	-0.5994	1.998

Table 1: Timing implied by the PowerPoint convention for the six right-going test-particle insertion events, using $\sigma_z = 0.6$ mm. The source centre column assumes an oncoming primary electron bunch centred at the IP at $ct = 0$.

This is an important distinction for OPALX comparisons. A deck that places all twelve witnesses at their different longitudinal positions at a single OPALX time $t = 0$, with the primary bunch also centred at the IP at $t = 0$, is a simultaneous spatial snapshot. It does not reproduce the PowerPoint insertion timing for all witnesses except the $ct = 0$ pair.

10.2 Slide-Timed OPALX Pair Decks

OPALX has one simulation clock per run. To reproduce the slide insertion timing without changing the numerical witness coordinates, the comparison now uses one electron–positron witness pair per OPALX input deck. For pair i , the witness `FROMFILE` rows retain the `TestParticleOrbit` values

$$(x, y, z, p_x, p_y, p_z)_{\text{OPALX}} = (x_i, y_i, s_i, P_{x,i}/m_e c, P_{y,i}/m_e c, P_{s,i}/m_e c),$$

while only the primary source offset is changed:

$$R_{0z,\text{source}} = s_{\text{IP}} - ct_i, \quad R_{0z,\text{witness}} = s_{\text{IP}}.$$

Thus at the first OPALX field evaluation,

$$s_i - s_c = 2ct_i,$$

which is the timing convention in Table 1. The generated decks are under the OPALX sandbox `timing_pairs/` directory; the generator is `generate_timing_pair_inputs.py`. These decks deliberately fix only the primary-bunch sampling time. They do not tune the field normalization, species mapping, source species, source direction, or pusher details.

Pair	ct_i/σ_z	$R_{0z,\text{source}}$ [mm]	$R_{0z,\text{witness}}$ [mm]	$s_i - s_c$ [mm]	$(s_i - s_c)/\sigma_z$
1	-1.500	3.9000	3.0000	-1.8000	-3.000
2	-0.999	3.5994	3.0000	-1.1988	-1.998
3	-0.501	3.3006	3.0000	-0.6012	-1.002
4	0.000	3.0000	3.0000	0.0000	0.000
5	0.501	2.6994	3.0000	0.6012	1.002
6	0.999	2.4006	3.0000	1.1988	1.998

Table 2: Generated OPALX source and witness offsets for the slide-timed pair decks. The metadata check gives $\max |(s_i - s_c)/\sigma_z - 2ct_i/\sigma_z| = 0$ to the printed precision.

All six slide-timed decks were run with $\Delta t = 1$ fs, $N_{\text{source}} = 400000$, a $32 \times 32 \times 64$ field mesh, and ten OPALX steps. The runs used `COPY_TIME=0`, so the witnesses see only the tracked primary source container. Every run reached the active BeamBeam state with three active containers, one source container and one witness in each witness container. Each run wrote one final H5 group at

$$t = 1.0 \times 10^{-14} \text{ s.}$$

The first-kick CSV diagnostic was enabled with `OPALX_BB_WITNESS_KICK_STEPS=1`. Each pair CSV contains four data rows: steps 0 and 1 for the electron and positron witness containers. The exact OPALX run command used for the six-pair serial validation was:

```
for pair in 1 2 3 4 5 6; do
  dir="/Users/adelman/git/opalx-beambeam/sandbox/track12particles/opalx/timing_pairs/pair${pair}"
  stem="track12_pair${pair}_slide_timed_one_source_1fs_400k_10steps"
  echo "running pair ${pair}"
  (cd "$dir" && env OMP_NUM_THREADS=8 \
    OPALX_BB_WITNESS_KICK_CSV="${stem}_kicks.csv" \
    OPALX_BB_WITNESS_KICK_STEPS=1 \
    /Users/adelman/git/opalx-beambeam/build_openmp/src/opalx "${stem}.in")
done
```

10.3 First-Kick Field Comparison

Table 3 compares the OPALX first-kick field magnitudes with the manufactured anisotropic Gaussian at the same witness positions and at the same slide-timed source offsets. The table uses the oncoming manufactured source convention implied by the PowerPoint timing sketch.

Pair	$s - s_{\text{IP}}$ [mm]	$ E _{\text{OPALX}}$ [V/m]	$ E _{\text{manuf.}}$ [V/m]	$ E _{\text{OPALX}}/ E _{\text{manuf.}}$	$ B _{\text{OPALX}}/ B _{\text{manuf.}}$
1	-0.899887	3.8787e7	5.3768e7	0.721	0.721
2	-0.599287	8.4742e8	6.5785e8	1.288	1.288
3	-0.300487	3.8295e9	2.9315e9	1.306	1.306
4	0.000113	6.2682e9	4.8444e9	1.294	1.294
5	0.300713	3.8666e9	2.9333e9	1.318	1.318
6	0.599513	8.3881e8	6.5864e8	1.274	1.274

Table 3: First-kick field magnitudes for the slide-timed pair decks, compared with the oncoming manufactured source. Electron and positron rows have the same field magnitudes for each pair, so the table lists the pair mean. Key result: the field magnitude disagreement is now at the 0.72–1.32 level, not orders of magnitude.

Repeating the same first-kick field evaluation with a copropagating manufactured source gives nearly the same field-magnitude ratios:

$$0.724 \leq |E|_{\text{OPALX}}/|E|_{\text{manuf.}} \leq 1.316, \quad 0.724 \leq |B|_{\text{OPALX}}/|B|_{\text{manuf.}} \leq 1.316.$$

The equality of the electric and magnetic ratios reflects that both models have $|B| \simeq |E|/c$ for this boosted-source geometry.

10.4 Ten-Step Final-State Comparison

The final ten-step phase-space comparison is summarized in Tables 4 and 5. Both tables use the H5 charge assignment for the OPALX witness containers: container 1 is the electron witness and container 2 is the positron witness. The reported values are the maximum absolute OPALX-minus-manufactured differences over the two witness species for each pair.

Pair	$\max \Delta x $ [μm]	$\max \Delta p_x $	$\max \Delta s $ [μm]
1	0.000467 (0,0%)	0.000626 (105,1%)	0.000000051 (0,0%)
2	0.005925 (0,3%)	0.007929 (109,2%)	0.000006665 (0,0%)
3	0.026389 (1,3%)	0.035290 (109,3%)	0.000131402 (0,0%)
4	0.043494 (2,2%)	0.058121 (109,3%)	0.000357354 (0,0%)
5	0.026306 (1,3%)	0.035082 (109,5%)	0.000130204 (0,0%)
6	0.005873 (0,3%)	0.007815 (109,2%)	0.000006535 (0,0%)

Table 4: Ten-step final-state differences for the slide-timed OPALX pair decks compared with an oncoming manufactured source. Percentages in parentheses are relative to the corresponding manufactured value and are rounded to one digit after the decimal comma. The largest differences are $\max |\Delta x| = 0.04349 \mu\text{m}$, $\max |\Delta p_x| = 0.05812$, and $\max |\Delta s| = 3.57 \times 10^{-4} \mu\text{m}$.

Pair	$\max \Delta x $ [μm]	$\max \Delta p_x $	$\max \Delta s $ [μm]
1	0.000054	0.000072	0.000000006
2	0.000885	0.001181	0.000000049
3	0.003981	0.005311	0.000000528
4	0.006549	0.008740	0.000001231
5	0.004011	0.005352	0.000000544
6	0.000884	0.001179	0.000000045

Table 5: Ten-step final-state differences for the same slide-timed OPALX pair decks compared with a copropagating manufactured source. The largest differences are $\max |\Delta x| = 0.00655 \mu\text{m}$, $\max |\Delta p_x| = 0.00874$, and $\max |\Delta s| = 1.23 \times 10^{-6} \mu\text{m}$.

Numerical conclusion: after fixing the slide timing, the current OPALX one-source decks agree much more closely with a copropagating manufactured source than with an oncoming manufactured source. This is a convention result, not a field or pusher tuning result. The slide-timed OPALX decks still use the existing tracked primary source momentum with positive p_z . Therefore the next physics decision is to explicitly choose the OPALX primary-source direction convention for the intended oncoming comparison before changing field normalization, species mapping, or Boris pusher details.

For the OPALX comparison, CAIN is no longer used as an active reference model. The same twelve numerical witness-particle initial conditions are read into OPALX through the two **FROMFILE** distributions and into the manufactured Python model. Table 6 compares the OPALX input files against the manufactured-script initial state. The reported deltas are OPALX minus manufactured. The agreement is exact for all listed coordinates and momenta.

Species	Pair	x [μm]	y [μm]	s [mm]	p_x	p_y	p_z	Δx [μm]	Δy [μm]	Δs [μm]	Δp_x	Δp_y	Δp_z
electron	1	1.9443250757	0	-0.9000	0	0	1.7320511804	0	0	0	0	0	0
electron	2	1.9443250757	0	-0.5994	0	0	1.7320511804	0	0	0	0	0	0
electron	3	1.9443250757	0	-0.3006	0	0	1.7320511804	0	0	0	0	0	0
electron	4	1.9443250757	0	0.0000	0	0	1.7320511804	0	0	0	0	0	0
electron	5	1.9443250757	0	0.3006	0	0	1.7320511804	0	0	0	0	0	0
electron	6	1.9443250757	0	0.5994	0	0	1.7320511804	0	0	0	0	0	0
positron	1	1.9443250757	0	-0.9000	0	0	1.7320511804	0	0	0	0	0	0
positron	2	1.9443250757	0	-0.5994	0	0	1.7320511804	0	0	0	0	0	0
positron	3	1.9443250757	0	-0.3006	0	0	1.7320511804	0	0	0	0	0	0
positron	4	1.9443250757	0	0.0000	0	0	1.7320511804	0	0	0	0	0	0
positron	5	1.9443250757	0	0.3006	0	0	1.7320511804	0	0	0	0	0	0
positron	6	1.9443250757	0	0.5994	0	0	1.7320511804	0	0	0	0	0	0

Table 6: Initial witness-particle phase space used by both OPALX and the manufactured Python model. The momenta are normalized OPALX momenta $p_i = \beta_i \gamma$. All deltas are OPALX minus manufactured.

The agreement is qualitative, not exact. This is expected: CAIN is a PIC code, while the manufactured model is a rigid continuum source. Differences can come from source disruption, finite macro-particle representation, exact timing and coordinate conventions, and from the fact that the manufactured model does not self-consistently evolve either colliding electron bunch.